

Day 17

Priority Encoder (8x3) in Verilog

Verilog Code

Priority Encoder code (8x3)

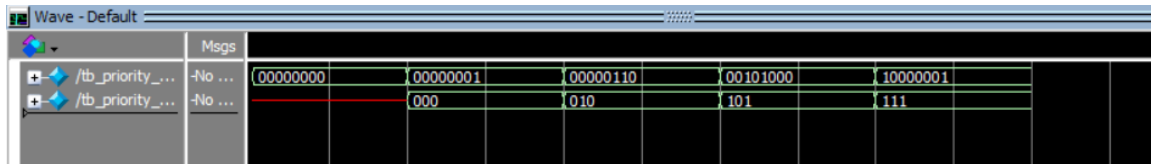
```
1 module priority_encoder(in,code);
2   input [7:0]in;
3   output reg [2:0]code;
4   always @(*) begin
5       if (in[7]) code = 3'b111;
6       else if (in[6]) code = 3'b110;
7       else if (in[5]) code = 3'b101;
8       else if (in[4]) code = 3'b100;
9       else if (in[3]) code = 3'b011;
10      else if (in[2]) code = 3'b010;
11      else if (in[1]) code = 3'b001;
12      else if (in[0]) code = 3'b000;
13      else code = 3'bxxx;
14   end
15 endmodule
```

Testbench Code

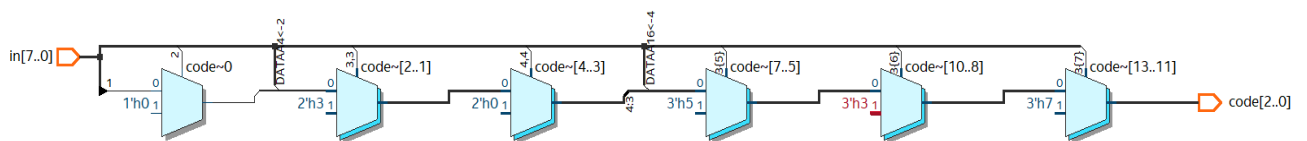
Priority Encoder Tb code (8x3)

```
1 module tb_priority_encoder;
2   reg [7:0] in;
3   wire [2:0] code;
4   priority_encoder dut (in,code);
5   initial begin
6       $monitor("in = %b code= %b",in,code);
7       in = 8'b00000000; #10;
8       in = 8'b00000001; #10;
9       in = 8'b00000110; #10;
10      in = 8'b00101000; #10;
11      in = 8'b10000001; #10;
12      $finish;
13   end
14 endmodule
```

Waveform



Schematic



EDA Tools Used

- IntelQuartusPrime
- ModelSim

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